When three output measurements identify channel purity

Research direction and responsibility · Kirtus Justus. Mathematical and computational assistance · ChatGPT (OpenAI). Port Coquitlam, British Columbia, Canada · 7 November 2026.

A fixed joint detector supplies a complete model phase curve and an extra-outcome rate f. Separate X, Y and Z measurements on the first output qubit use a product input $|00\rangle$. What follows if all three probabilities equal their minimum-purity reference values? The answer changes at the upper endpoint of f.

> For every $0 \le f < r$, at least two distinct purities fit the reference triple. At $f=0$ the full interval [0.29284162828..., 1] fits. At $f=r$ the reference triple fixes purity to 0.501235795572884... . These are population-level implications of the specified trace-preserving model.

What was supplied and what was checked

Use the normalized Choi ket $|M\rangle = \mathrm{vec}(M)/2$, output indices first. Put $p=|I_4\rangle$, $u=|iZ\otimes I\rangle$ and $v=|iX\otimes I\rangle$. With $s=131/320$, $h=0.99$, $q=81711338569043/224995905216000$, set $k=(sh-\sqrt{[shq]})^2/[sh(1-h)]$, $d=s+k$, $r=1-d$. The specified lower-endpoint phase curve fixes the p,u compression to $|w\rangle\langle w|$, $w=\sqrt{s}\,p-\sqrt{k}\,u$. Write $\eta=w/\sqrt{d}$, $\theta=(\sqrt{k}\,p+\sqrt{s}\,u)/\sqrt{d}$, and D for the projection perpendicular to p,u,v. The θ direction has zero population. The remaining populations on η,v,D are $d, f, \ell = r-f$.

The minimum state $\rho_\star = d|\eta\rangle\langle\eta| + f|v\rangle\langle v| + \ell D/13$ has purity $J(f) = d^2+f^2+\ell^2/13$. Its probabilities are $pZ_\star = d + 6\ell/13$ and $pX_\star = pY_\star = 1/2$. Write $\Delta j = pj - pj_\star$. The supplied global lower bound is $J(f) + (91/1230)\Delta Z^2 + (7/97)(\Delta X^2 + \Delta Y^2)$. I reconstructed its coefficients exactly over the rationals in a rotated frame. At $f=r/2$, the stated shifts (0.005, 0.007, −0.004) in lab order Z,X,Y give a positive trace-preserving state with purity 0.336263620116203 to the precision shown.

Observed f	Matching all three reference probabilities
0	Every purity in [J(0),1] occurs. An explicit pure unitary is on p. 3.
$0<f<r$	Some ambiguity always survives. At $f \approx 0.075590554093$ the full interval [J(f),1] occurs.
r	The only compatible state is $\rho_\star$ at the reference triple. Away from it the transverse component gives purity exactly.

The normalized input marginal and transfer of the same channel to $|00\rangle$ must hold to apply any of these results. The physical gate path after the reported history 111 was not measured in the supplied packet.

Supplied three-axis proof, sections 2–4. Source ↗

Keith et al. (2018), Joint quantum-state and measurement tomography with incomplete measurements. Source ↗

Projection on the actual support

Positive states with zero population in a direction annihilate it. The allowed linear perturbations therefore change at f=0 and f=r.

Define $A=\sqrt{(s/d)}$, $B=\sqrt{(k/d)}$, so $A^2+B^2=1$. The rotated transverse deviations are $\Delta_1=A\Delta X+B\Delta Y$ and $\Delta_2=-B\Delta X+A\Delta Y$. To see the rotation explicitly, write $A=\cos\alpha$, $B=\sin\alpha$, $W=\exp(-i\alpha Z)\otimes I$ and $T=\exp(-i\alpha Z/2)\otimes I$. The basis change $M\mapsto TW\dagger MT\dagger$ maps η,θ,v to $|I_4\rangle$, $|iZ\otimes I\rangle$, $|iX\otimes I\rangle$. It fixes the product input up to phase and rotates its first-qubit Bloch vector by $-\alpha$. This is a change of coordinates in the proof.

Support	Gram diagonal in order Z, Δ_1, Δ_2	Rank
0<f<r	1230/91, 97/7, 97/7	3
f=0	159/13, 25/2, 254/21	3
f=r	0, 0, 1/2	1

At f=0, take the reciprocals to get the globally valid excess bound $(13/159)\Delta Z^2+(2/25)\Delta_1^2+(21/254)\Delta_2^2$. This is locally sharp in all three directions. At f=r only Δ_2 can vary: $pZ=d$, $\Delta_1=0$ and $|\Delta_2|\le\sqrt{(dr)}$. In the surviving two-dimensional Choi block the diagonals are d,r and trace preservation makes the coherence γ real. Applying the channel to $|00\rangle$ gives $\Delta_2=\gamma$. Thus purity is exactly $d^2+r^2+2\Delta_2^2$. Every allowed γ gives a positive normalized Choi state.

Gram calculation

For each support S, compress each half-Pauli observable $2(Q\otimes I)_{out}\otimes|00\rangle\langle 00|_{ref}$ to S, then remove its components along the fixed population projections. Using half-Paulis gives the same probability deviations because the reference marginal is fixed. Call the resulting observable K and its output partial trace L. To impose zero marginal, project away $S(I\otimes H)S$ for traceless Hermitian H. The companion constructs the exact 15×15 Gram matrix on the traceless Pauli basis and uses its Moore–Penrose inverse.

For the interior, the Gram matrix on H is $(7/2)I$ in an orthonormal basis. At f=0 the two missing directions give the form $3\mathrm{Tr}(HG)+(1/8)\mathrm{Tr}(Y_1H)\mathrm{Tr}(Y_1G)$, with $Y_1=Y\otimes I$. The raw half-Pauli norms squared are 159/13,25/2,25/2; their reference partial traces have squared norms 0,0,5/4 with $|\mathrm{Tr}(Y_1L)|=1$ on the last axis. Subtracting 5/12−1/84 from the last entry gives 254/21. All transverse cross entries vanish in this frame.

For nonsingular Gram G, the equality candidate is $\rho^\star$ plus the sum of the projected observables with coefficients $G^{-1}\Delta$. This state meets all linear constraints. The lower bound is attained exactly when this candidate is positive semidefinite. At an interior point of its support small changes preserve positivity. For distant triples the certificate alone does not determine the optimum.

Which purities fit the reference triple?

The following constructions preserve all three output rates, f, the fixed-detector phase curve and both normalized Choi marginals. Work in the rotated coordinates of p. 2.

A nonzero interval for every f<r

The unresolved thirteen two-qubit Pauli directions contain six that preserve the first bit of $|00\rangle$ and seven that flip it. Keep the d and f weights on η and v. Starting with $\rho_\star$, put the six preserving weights entirely on $|I\otimes X\rangle$ and the seven flipping weights on $|X\otimes X\rangle$. Every component gives a computational-basis first-qubit output. The same three reference probabilities follow, and the state is mixed-unitary. Its purity is $J(f)+72\ell^2/169$. Mixing with $\rho_\star$ realizes every value over that interval, since $\mathrm{Tr}(\rho\rho_\star)=J(f)$. Its width vanishes only at f=r. This proves ambiguity at every positive residual without assuming the top endpoint is one.

A pure channel at the lower endpoint

At f=0 put $c_0=\sqrt{(d+6r/13)}$, $c_1=2\sqrt{d}-c_0$. On ordered pairs (00,11) and (01,10) use the unitary blocks $c_jI+i\sqrt{(1-c_j^2)}X$. Both c_j lie in $[-1,1]$. The total trace divided by four is $\sqrt{d}$, and the coefficients on $Z\otimes I$ and $X\otimes I$ vanish. On $|00\rangle$ the components carry distinct second-qubit flags, so the first-qubit density is diagonal with Z-positive rate $c_0^2=pZ_\star$. This pure Choi state and $\rho_\star$ agree on all specified outcomes. Their overlap is J(0); their mixtures cover [J(0),1].

An explicit interior example

Let f be the unique root in (0,r) of $df=(42/169)(r-f)^2$. Then $f\approx0.075590554093$. Put $U=(\sqrt{d}\,I+i\sqrt{f}\,X)/\sqrt{(d+f)}$, a first-qubit unitary. Choose a Hermitian involution N taking $|0\rangle$ to $U\dagger[\sqrt{(6/13)}|0\rangle-i\sqrt{(7/13)}|1\rangle]$ up to phase. Make the first entry a real by a phase choice and call the second entry b; the matrix with rows (a,b*) and (b,−a) has this property. The two-qubit unitary $V=(U\otimes I)[\sqrt{(d+f)}I_4+i\sqrt{\ell}(N\otimes X)]$ has the required visible Choi component.

Its two product-output branches have orthogonal second-qubit flags. The first branch gives first-qubit off-diagonal $-i\sqrt{(df)}$; the second gives $+i\ell\sqrt{42/13}$. The root condition cancels them. The Z-positive weight is $d+6\ell/13$. This proves all purities through one fit the reference triple at this interior f. The construction supplies no general pure endpoint for other values of f.

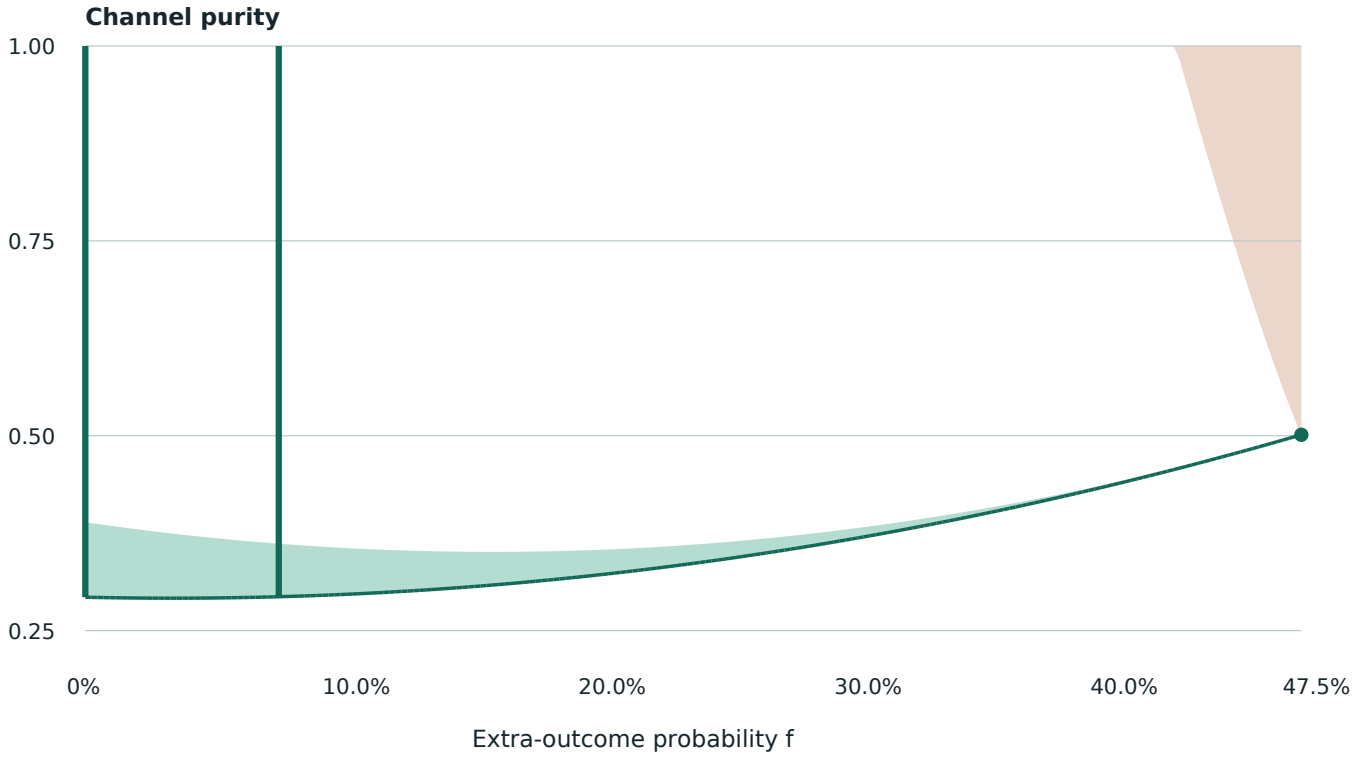
Near the upper boundary the opposite can happen. For example at $\ell=0.01$ and the reference triple, the preceding transverse bound gives purity ≤0.58577 (rounded upward), while the constructed sign-varying family at $\ell=0.01$ spans 0.9802 to 1 for a different first-qubit output. The data vectors must be specified to compare purity ranges.

For that upper check the reference transverse expectation is zero. With $\beta=2\sqrt{[3(1-\ell)\ell]}+4\ell$, the channel bound gives $d^2+f^2+(\beta^2+\ell^2)/2+2(1-\ell)\ell+\ell^2$ as one valid upper estimate. This is a conservative exclusion of purity one for the indicated data.

Transverse readout and channel purity near a support boundary, p. 2. Source ↗

Purities compatible with the three reference rates

This plot applies to one curve of observations: pX=pY=1/2, pZ=d+6(r−f)/13, and the same complete fixed-detector phase curve.



Green shows values with explicit channels. The vertical green lines give the full range at f=0 and f≈7.5591%. The dot gives the unique value at f=r. The rust region is excluded by the conservative transverse estimate; all values below the green lower curve are excluded by J. Other white values above the curve are unclassified by these constructions.

Using the bound with finite observations

The three Pauli components require distinct measurement settings on comparably prepared outputs. A valid confidence statement needs one simultaneous region for the underlying probabilities and calibrated inputs. Minimize the applicable certificate over that region and all compatible models. A confidence interval merely touching f=0 or f=r does not prove exact zero population, so using a boundary coefficient for the entire interval is unwarranted. Uncertainty in the detector or phase curve also releases the exact support constraint unless separately bounded.

White et al. (2022), Non-Markovian quantum process tomography. Source ↗